

School of Computer and Systems Sciences
Jawaharlal Nehru University
New Delhi
Mid-semester Examination- II

CS 406 - Numerical Methods
MCA 1st Semester
27 November 2024

Time: 2 hrs

Max. Marks: 20

Q 1. Use Newton's forward difference formula to find the unique polynomial $P(x)$ such that $P(1) = 1$, $P(2) = 0$, $P(3) = 27$ and $P(4) = 64$ (5)

$$= 3x^3 + 23x^2 - 21x + 1$$

Q 2. Compute the following integral using Trapezoidal rule for $p = 1$, and number of points = 3 (5)

$$I_p = \int_0^1 \frac{x^p dx}{x^3 + 10}$$

$$0.0478$$

Q 3. Compute approximation to $y(0.6)$ (4 decimals) of the IVP (10)

$$y' = x(x+y) - 2$$

$$y(0) = 2$$

Using Euler's method with step size $h = 0.2$

$$1.00576$$